1) Compare Type-1 and Type-2 Hypervisors for a Cloud Data Center Hosting Mission-Critical Applications.

Feature	Type-1 Hypervisor (Bare metal)	Type-2 Hypervisor (Hosted)
Installation	Runs directly on physical hardware	Runs on top of a host operating system
Performance	Very high, low overhead	Lower due to host OS overhead
Security	more Secure because there is not host OS	Less Secure
Manageability	Centralized management, ideal for cloud data centers	Easy to install but difficult to manage at large Scale
Examples	VMware ESXI, Microsoft Hyper-V Xen	Oracle VirtualBox, VMware Workstation

Analysis:-

Performance:-

* Type-1 Hypervisors Communicate directly with hardware, reducing latency and improving CPU, memory and I/O performance.

* Type-2 Hypervisors depend on the host operating system, Causing additional overhead and slower Performance..

Security:-

* Type-1 hypervisors Provide Stronger isolation between virtual machines and reduce the attrack surface.

* Type-2 hypervisors are more vulnerable because if the host operating System is Compromised, all virtual machines may also be affected

Justification:-

Type-1 Hypervisors is the better choice because it offers:

* High Performance with minimal overhead
* Strong security through direct hardware access and better VM isolation.

Manageability :-

* Type-1 hypervisors support centralized management, live migration, snapshots, making them suitable for enterprise cloud environment.

* Type-2 hypervisors are mainly intended for testing, development and personal use, not large-scale could data centers.

Conclusion :-

* Type-1 Hypervisor is the preferred solution for mission-critical cloud applications because it provides better performance, stronger security, and easier management compared to type-2 Hypervisors.

2) * A cloud Provider can securely run finance, healthcare and student portal applications on the same physical server by using virtualization architecture. Each application runs inside a separate VM with it's own operating system, CPU, memory storage. This ensures that one workload can't directly access or interfere with another.

How Virtualization supports isolation :-

1. Virtual Machine (VM) isolation :-

* Each workload runs in a separate VM

2. Resource isolation :-

* The hypervisor allocates dedicated CPU, memory, storage and network resources to each VM.

3. Network isolation :-

* Virtual networks and VLANS separate traffic between finance, healthcare, and student portal workloads.

Two Major Attack Surfaces

1. Hypervisor Attack :-

* If an attacker exploits a vulnerability in the hypervisor, multiple VMs on the same host could be compromised.

2. VM Escape Attack:

* A malicious program inside one VM escapes its virtual environment and gains access to the host system or other VMs.

Conclusion :-

* Virtualization architecture provides strong isolation by separating workloads into independent virtual machines, allocating dedicated resources, and isolating network traffic. However, cloud providers must protect against hypervisor attacks and VM escape attacks by keeping hypervisors updated, applying security patches, and continuously monitoring the environment

3) The given VM host utilization indicates that the cloud server is under heavy load, resulting in poor performance and slow virtual machine startup.

Analysis of the Bottlenecks

1. CPU utilization - 92%

* CPU usage is extremely high.
* Bottleneck: CPU resource shortage.

2. Memory utilization - 88%

* Memory usage is close to maximum capacity.
* Bottleneck: insufficient RAM

3. Storage I/O Latency - 35 ms

* A storage latency of 35 ms is high
* Bottleneck: slow storage subsystem or overloaded disks.

4. Average VM Boot time - 170 Seconds
cause

* A VM boot time of 170 seconds is much longer than normal

* This is caused by high CPU usage, insufficient memory and slow storage performance.

* Bottleneck: Combined effect of CPU, memory, and storage limitations

Recommended Corrective Actions

1. Reduce CPU Load

* upgrade CPU resources or add more host servers.

* Balance workloads using load balancing or VM migration

2. Increase memory:

* Add more RAM to the host

* Optimize VM memory allocation and remove unused VMs

Conclusion :-

* The VM host suffers from CPU overload (92 %), high memory utilization (88 %), and high storage (I/O) latency (35 ms), leading to a slow VM boot time of 170 seconds

4)

* A Software Defined Data Center (SDDC) uses software-based control and virtualization to manage compute, storage and networking resources.

1. Compute Virtualization :-

* In a traditional data center, applications are usually tied to physical servers

* SDDC uses virtual machines (VMs) to run multiple workloads on the same physical server

Result: Faster deployment and better utilization of Computing resources.

2. Storage Virtualization:

* Traditional data centers often depend on fixed physical storage systems.

* SDDC abstracts physical storage and presents it as virtual storage resources.

Result: Faster storage provisioning and easier scalability.

3. Network virtualization:

* Traditional networks require manual configuration of physical switches and routers.

* Network policies can also be applied centrally.

Result: Faster network configuration, improved flexibility, and easier management.

SDDC vs Traditional Data Center

Feature	Traditional Data Center	SDDC
Compute	physical servers	Virtual machines
Storage	physical storage devices	virtualized storage
Network	hardware - based Configuration	Software defined networking
Provisioning	Mostly manual	Software - Controlled
scalability	slower	Higher

Conclusion:-

* SDDC improves data Center agility by abstracting Compute, storage and network resources from their physical hardware. Therefore SDDC is more flexable and scalable than a traditional data center.

* Multi-cloud federation allows different cloud providers to work together and share resources and services.

1. Analysis of the identity mismatch:-

* Different cloud providers may use different user IDs and identity formats.
* Authentication systems may use different protocols

2. Secure identity Design using federation.

User → Identity provider (idp) → cloud providers

* Establish a trusted identity provider (idp)
* Users authenticate with the idp instead of separately authenticating with every cloud provider.

3. Privacy and Security Measures.

* Use strong authentication and MFA
* Apply least - privilege access to users receive only necessary permissions.
* Encrypt identity information during transmission
* Maintain audit logs for authentication and access activities
* Regularly review and revoke unnecessary permissions.

Conclusion:-

* The identity mismatch can be solved by implementing a federated identity architecture with a trusted identity provider, standardized authentication, role mapping, and secure authorization policies